Conclusions

Large passive reflectors are extremely practical. They can be used to replace active repeaters on line-of-sight microwave systems and can also be used on diffraction and tropospheric scatter systems. By the use of high power transmitters and very sensitive receivers long, two or more hop systems may be effectively utilized. The space loss on a two hop system is *independent* of frequency and on a three or more hop system is *dependent* on frequency with *less* loss at the *higher* frequencies. By the use of frequencies of several gigacycles with their higher gain antennas, the system losses can be quite low. Large terminal antennas and passive reflectors may be sited and installed using the same engineering and installation practices normally used for tropospheric scatter systems. Large passive reflectors are also economical. When used to replace active microwave repeaters or to extend a tropospheric scatter or diffraction system considerable savings in initial and operating cost can be realized. The reliability of communication systems using large passive reflectors will be considerably higher due to a reduction of the quantity and/or complexity of the electronic equipment.

Large passive reflectors should be considered as another tool which can be used by the microwave engineer for planning and engineering communication systems. When used effectively with line-of-sight, diffraction and the tropospheric scatter modes of propagation, communication systems may be engineered more economically, with more reliability and with a decrease in the operating and maintenance problems.

An Analog Multiplier Using Two Field Effect Transistors*

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Summary—**A four-quadrant analog multiplier is described which uses the variable channel conductance properties of the field effect transistor. This multiplier is characterized by moderate accuracy, high speed and circuit simplicity. As such, it may be useful in many applications where speed is important but a high degree of accuracy is not. Such an application is found in synchronous detection circuits desirable in some types of communication systems.**

Introduction

THE ANALOG multiplier is a device whose output is a quantity which is proportional to the instantaneous product of two time-varying input quantities. Such a device is finding wide application in the present day art, particularly in the fields of analog computers and communication systems. Multipliers for analog computer use are generally characterized by very high accuracy and moderate speed requirements. On the other hand, multipliers for communication systems (for example, synchronous detectors) require only moderate accuracy but must have a relatively large bandwidth.

There are many ways of practically implementing an analog multiplier. Among these are mechanical servo multipliers[1] and Hall-effect multipliers,[2] both of which are limited in their bandwidth capabilities with respect to at least one of the input quantities. In order to obtain high speed multipliers, more complicated transistor or vacuum tube circuits have been developed.[1,3,4] For instance, one successful approach involves a pulse train which is amplitude modulated by one of the input signals, and pulse-width modulated by the other input signal. The pulse train is then integrated, the integral being the desired product.

This paper describes a four quadrant analog multiplier suitable for use in communication system applications. It makes use of the variable channel conductance properties of the field effect transistor, and is characterized by moderate accuracy, wide bandwidth and simple circuitry.

The Field Effect Transistor

A planar field effect transistor is shown in Fig. 1 (this is only one of the many possible configurations). It is a three layer semiconductor device in which the two outside layers are heavily doped as one type of extrinsic semiconductor (*P* type in this example), and the middle layer is lightly doped with an opposite type (donor) impurity. The two outside layers are connected together and become the control element, denoted the *gate*. The main path for current flow is through the middle layer (*channel*). The two connections to the channel are called the *source*

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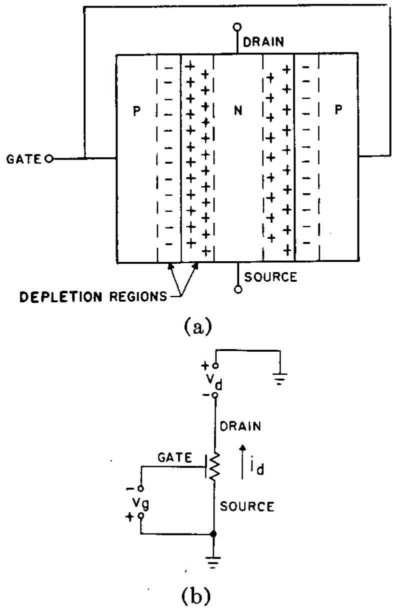

Fig. 1—The field effect transistor. (a) Planar configuration. (b) Schematic representation.

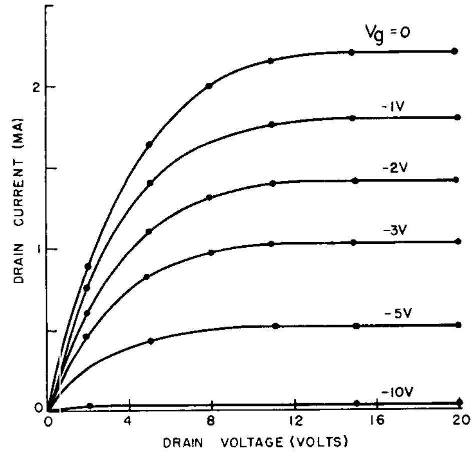

Fig. 2—V-1 Characteristics of a crystalonics field effect transistor, C612.

and the *drain.* The schematic representation for this device is also shown in Fig. 1.

If a voltage is applied from gate to source in such a way as to back bias the gate junctions, then a depletion region will be formed about each gate junction. This depletion region will extend primarily into the channel, since the gate regions are heavily doped with respect to the channel. Since no free charge carriers can exist in the depletion regions due to the high fields, the effective cross-sectional area and thus the channel conductance, are reduced by the existence of the depletion region.

It is therefore seen that the channel resistance will vary with the magnitude of the gate voltage; *i.e.*, as the gate voltage is increased in magnitude, the width of the depletion region increases, the channel width decreases, and the channel resistance increases.

For a given gate voltage and for small drain voltages (assuming the source to be grounded), the channel then looks like a resistor of some value. However, as the drain voltage is increased in such a direction as to further reverse bias the gate junctions, the width of the depletion region at the drain increases. It will finally become wide enough to reduce the channel width to zero at the drain, and the channel becomes a constant current source. This condition is called pinchoff. Pinchoff will occur at a lower drain voltage as the gate voltage is increased.

These characteristics are summarized in a plot of typical field effect characteristics shown in Fig. 2. Note that, because of the high input and output impedances (in pinchoff), the field effect transistor bears a close similarity to the vacuum tube pentode. The main difference occurs for low drain voltages, where the field effect transistor exhibits the variable channel conductance effect previously described. It is this characteristic in which we are interested.

The channel will look like a good resistor for a given gate voltage for either positive or negative drain voltages, as long as the drain voltages are small enough so that they do not significantly disturb the depletion regions set up by the gate bias. Thus for purposes herein we will consider the drain as being unbiased, with a bipolar signal applied to it. The gate junctions will be reversed biased sufficiently to insure that the perturbation of the depletion region by nonzero drain voltages may be neglected.

Shockley[5] has shown that the channel conductance for the case of zero drain voltage should obey the relation

$$G = G_0\left[1 - \left(\frac{V_g}{V_0}\right)^{1/2}\right],$$

where

G = channel conductance,
V_g = gate to source voltage,
V_0 = value of V_G required to pinch off the channel, and
G_0 = value of G for $V_g = 0$.

However, practical field effect transistors seem to obey a relation which is linear over wide ranges of gate voltage. As an example, the channel conductance vs gate voltage characteristic for a commercially available field effect transistor (Crystalonics type C-612) is shown in Fig. 3 under the condition of zero drain voltage. We will therefore model the channel conductance dependence on the gate voltage by the relation

$$G = G_0\left(1 - \frac{V_g}{V_0}\right). \tag{1}$$

Using this simple model of the field effect transistor, a simple analog multiplier will be described and experimental results given. Finally, the effects of a nonlinear dependence of the channel conductance on gate voltage and the dependence of this conductance on drain voltage is discussed in the Appendix. It is shown there that these effects may be balanced out to a reasonable degree by the circuit to be described.

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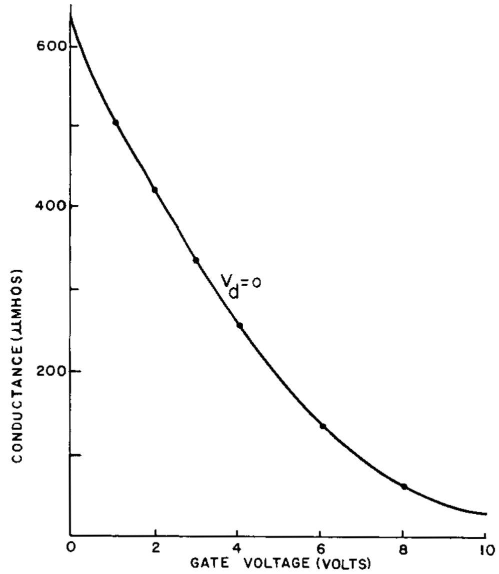

Fig. 3—AC conductance characteristics of Crystalonics field effect transistor, C612.

A Simple Multiplier

Consider the circuit of Fig. 4, and assume that the simple linear model of (1) holds for the field effect transistor. Then the channel conductance is given by

$$G = G_0\left(1 - \frac{V_{gg} + e_2}{V_0}\right) = G_0\left(1 - \frac{V_{gg}}{V_0}\right) - G_0\frac{e_2}{V_0}.$$

The drain current is

$$i_d = Gke_1 = G_0k\left(1 - \frac{V_{gg}}{V_0}\right)e_1 - G_0k\frac{e_1e_2}{V_0},$$

and the output current i_0 is then

$$i_0 = G_0k\left(1 - \frac{V_{gg}}{V_0}\right)e_1 - \frac{k}{R}e_1 - G_0k\frac{e_1e_2}{V_0}.$$

If R is chosen so that its conductance is equal to the channel conductance for $e_2 = 0$, that is,

$$R = \frac{1}{G_0\left(1 - \frac{V_{gg}}{V_0}\right)},$$

then

$$i_0 = -k\frac{G_0}{V_g}e_1e_2,$$

and i_0 is proportional to the product of the two input variables. Thus Fig. 4 represents a simple analog multiplier. The input powers required are small (the gate input power is essentially zero), power gain may be realized (a load resistance small compared to the channel conductance may be driven), and the frequency response is limited only by the response of the field effect transistor and transformer. Commerically available field effect transistors have cutoff frequencies of several megacycles. If the bandwidth is too wide for a reasonable transformer to handle, then electronic phase conversion may be used.

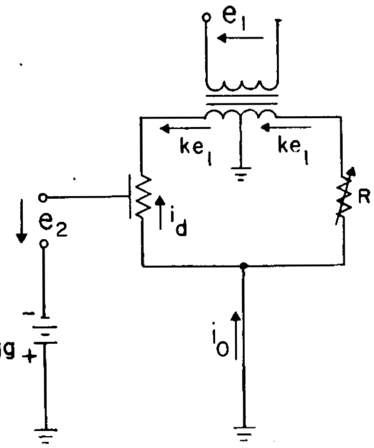

Fig. 4—Analog multiplier using one linear field effect transistor.

A Balanced Analog Multiplier

The simple multiplier just described suffers from the fact that practical field effect transistor nonlinearities cause additional undesired terms to appear in the output. By using two field effect transistors in a balanced configuration, it is possible to minimize this distortion to a great extent.

A balanced four quadrant field effect transistor analog multiplier is shown in Fig. 5. The two potentiometers serve as an effective variable center tap for each of the transformers, and are used to balance out the distortion terms. The minimization of the distortion terms are considered in more detail in the Appendix, where it is shown that second order nonlinearities in the channel conductance versus gate voltage characteristics may be balanced out, but distortion terms due to the dependence of the channel conductance on drain voltage may only be minimized.

For present purposes of analysis, assume that the field effect transistors in Fig. 5 are described by the linear model of (1), but that they have different values for the parameters G_0 and V_0. Assume the turns ratios of the transformers are each unity for simplicity (this does not affect the analysis).

Then

$$V_{g_1} = V_{gg} + k_g e_2$$

$$V_{g_2} = V_{gg} - (1 - k_g)e_2$$

$$V_{d_1} = k_d e_1$$

$$V_{d_2} = (1 - k_d)e_1,$$

where k_g and k_d are determined by the setting of the potentiometers associated with e_2 and e_1, respectively, and range from zero to one.

From (1), the load current is

$$i_0 = G_1V_{d_1} + V_{d_2}G_2 = G_{01}\left[1 - \frac{V_{gg} + k_ge_2}{V_{01}}\right]k_de_1 - G_{02}\left[1 - \frac{V_{gg} - (1 - k_g)e_2}{V_{02}}\right](1 - k_d)e_1,$$

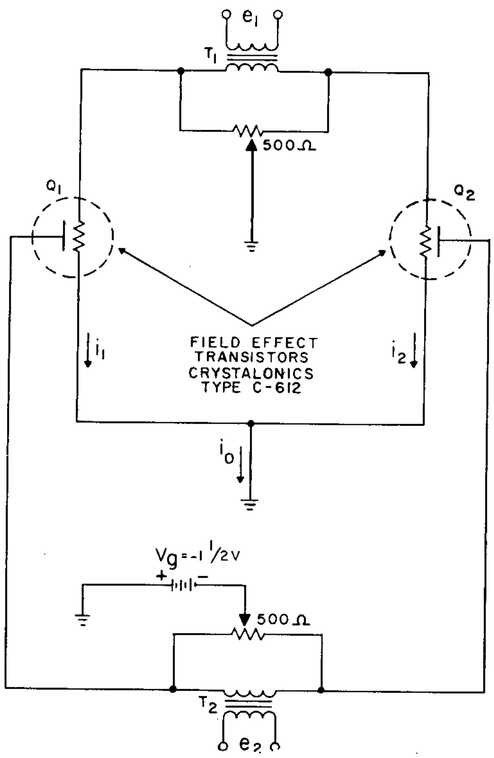

Fig. 5—Schematic diagram of the analog multiplier.

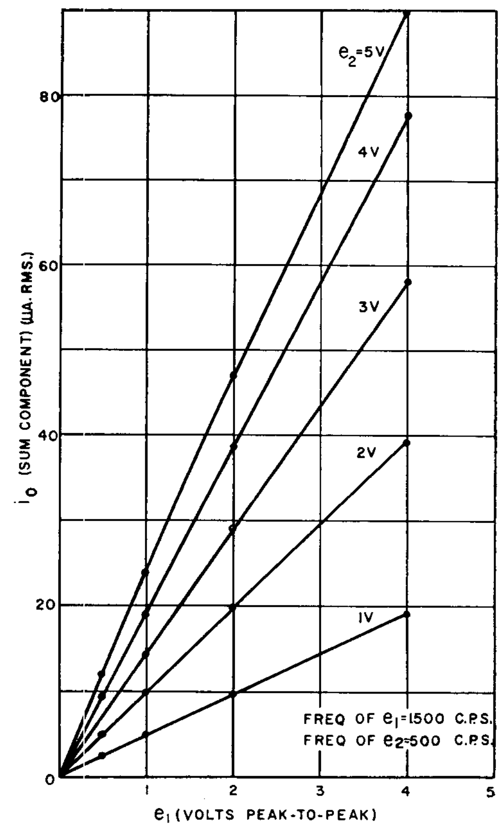

Fig. 6—Producting characteristics.

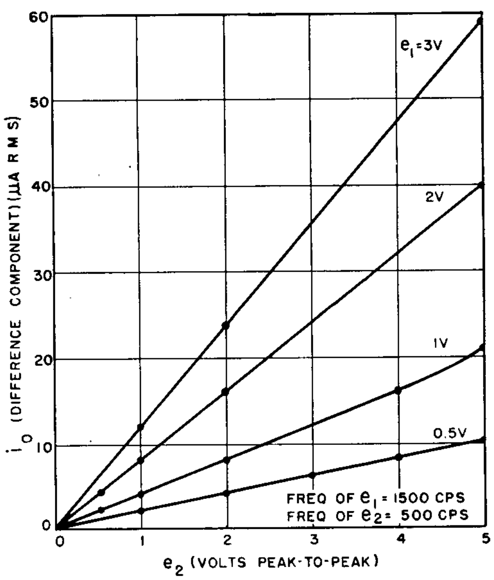

Fig. 7—Producting characteristics.

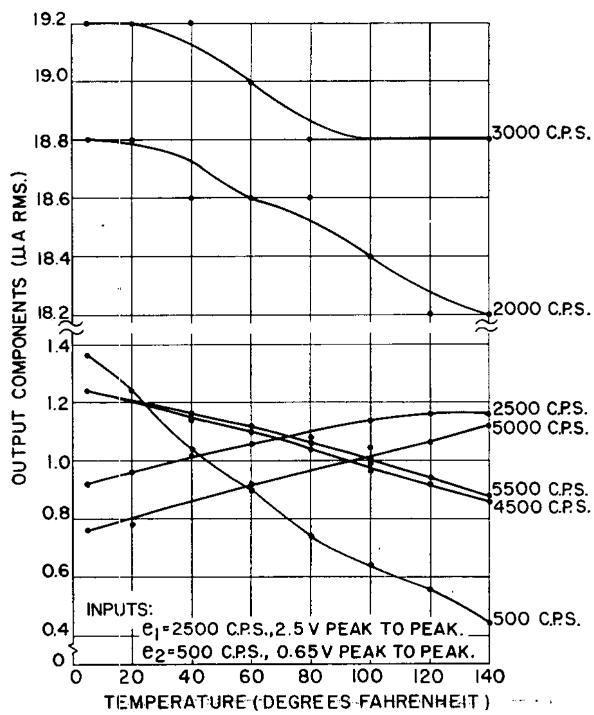

Fig. 8—Temperature characteristics.

where G_{01}, V_{01}, G_{02} and V_{02} are the appropriate parameters for the transistors Q_1 and Q_2. This may be written as

$$i_0 = -\left[\frac{G_{01}k_g k_d}{V_{01}} + \frac{G_{02}(1-k_g)(1-k_d)}{V_{02}}\right]e_1e_2 + \left[G_{01}\left(1-\frac{V_{gg}}{V_{01}}\right)k_d - G_{02}\left(1-\frac{V_{gg}}{V_{02}}\right)(1-k_d)\right]e_1.$$

The first term describes the desired operation of the analog multiplier. The undesired second term may be eliminated by adjusting the drain potentiometer so that

$$k_d = \frac{G_{02}\left(1-\frac{V_{gg}}{V_{02}}\right)}{G_{01}\left(1-\frac{V_{gg}}{V_{01}}\right) + G_{02}\left(1-\frac{V_{gg}}{V_{02}}\right)}.$$

For identical transistors, $k_d = \frac{1}{2}$ (a center tapped transformer). It is seen that the gate potentiometer is not needed for linear elements. However, it is shown in the Appendix that the gate potentiometer is useful in reducing distortion terms due to nonlinearities.

Experimental Results

The circuit of Fig. 5 was constructed and tested using a pair of Crystalonics-type C-612 field effect transistors. The intended use of the circuit was for low frequencies (300 to 3000 cps), and the transformers were chosen accordingly. The frequency response of this circuit was therefore limited by the transformer characteristics.

Typical characteristics are shown in Figs. 6 and 7 These were obtained by feeding one input with a 1500-cps signal and the other with a 500-cps signal. The output current should then consist of two components, a sum component at 2000 cps and a difference component at 1000 cps. The amplitudes of these components should be equal and proportional to the product of the input amplitudes. In Fig. 6 is plotted the sum component as a function of e_1, with e_2 as a parameter. In Fig. 7 is plotted the difference component as a function of e_2, with e_1 as a parameter. The output components were determined by making measurements with a wave analyzer on the voltage developed across a small load resistor. It is seen from these curves that this multiplier is linear over a range of roughly 4 v peak-to-peak at either input (the transformers used had a turns ratio of one-to-one).

In Fig. 8 is shown the temperature dependence of the multiplier. The inputs were sinusoids, e_1 being at a frequency of 2500 cps and an amplitude of 2.5 v peak-to-peak, and e_2 being at a frequency of 500 cps and an amplitude of 0.65 v peak-to-peak. Because silicon devices were used, the temperature of operation is not critical for normally encountered conditions. Also shown in Fig. 8 is a detailed measurement of the distortion terms encountered with this particular set of input conditions. The frequencies of these distortion terms agree with the analysis given in the Appendix.

Conclusion

A four quadrant analog multiplier using a pair of field effect transistors has been described. This approach to analog multiplication is characterized by moderate accuracy, high speed and circuit simplicity. Depending upon advances in the art with respect to the field effect transistor, in particular with respect to reproducibility, linearity and cost, this sort of multiplier may find use in many applications, communications and otherwise in the near future.

Appendix

In order to obtain a more exact analysis of the circuit of Fig. 1, it is necessary to assume a better model of the channel conductance than provided in (1). It is also necessary to consider nonlinear drain voltage-current characteristics.

Assume that the expression for the instantaneous drain current in either transistor is

$$i_d = G(V_d + CV_d^2), \tag{2}$$

where V_g is the instantaneous drain voltage and G is the instantaneous channel conductance, which will now be taken as

$$G = G_0(1 + AV_g + BV_g^2), \tag{3}$$

where V_g is the instantaneous gate voltage. G_0 is the value of G for $V_g = 0$, and A and B are constants, dependent on the bulk semiconductor properties. Substituting (3) into (2), one obtains

$$i_d = G_0(1 + AV_g + BV_g^2)(V_d + CV_d^2).$$

Letting $V_g = V_{gg} + v_g$ and V_d [6] $= V_{da} + v_d$, where V_{gg} and V_{dd} are the dc bias voltages for the gate and drain, respectively, and v_g and v_d are the signals at the gate and drain, respectively, we have

$$\begin{aligned} i_d &= G_0[1 + A(V_{gg} + v_g) + B(V_{gg} + v_g)^2] \\ &\quad \cdot[(V_{dd} + v_d) + C(V_{dd} + v_d)^2] \\ &= G_0[(1 + AV_{gg} + BV_{gg}^2) + (A + 2BV_{gg})v_g + Bv_g^2] \\ &\quad \cdot[(V_{dd} + CV_{dd}^2) + (1 + 2CV_{dd})v_d + Cv_d^2] \\ &= G_0[V_{dd}(1 + AV_{gg} + BV_{gg}^2)(1 + CV_{dd}) \\ &\quad + V_{dd}(1 + CV_{dd})(Av_g + 2BV_{gg}v_g + Bv_g^2) \\ &\quad + (1 + AV_{gg} + BV_{gg}^2)(1 + 2CV_{dd})v_d \\ &\quad + (A + 2BV_{gg})(1 + 2CV_{dd})v_gv_d \\ &\quad + B(1 + 2CV_{dd})v_g^2v_d + C(1 + AV_{gg} + BV_{gg}^2)v_d^2 \\ &\quad + C(A + 2BV_{gg})v_gv_d^2 + BCv_g^2v_d^2]. \end{aligned}$$

[6] A nonzero drain bias voltage is considered here. This is noteworthy, since the earlier analysis assumed a zero dc drain voltage.

Using an additional subscript, 1 and 2, for the terms corresponding to transistors Q_1 and Q_2, respectively, we have

$$\begin{aligned} i_1 = G_0[&V_{dd}(1 + A_1V_{gg} + B_1V_{gg}^2)(1 + C_1V_{dd}) \\ &+ V_{dd}(1 + C_1V_{dd})(A_1v_{g1} + 2B_1V_{gg}v_{g1} + B_1v_{g1}^2) \\ &+ (1 + A_1V_{gg} + B_1V_{gg}^2)(1 + 2C_1V_{dd}v_{d1}) \\ &+ (A_1 + 2B_1V_{gg})(1 + 2C_1V_{dd}v_{g1}v_{d1}) \\ &+ B_1(1 + 2C_1V_{dd})v_{g1}^2v_{d1} + C_1(1 + A_1V_{gg} + B_1V_{gg}^2)v_{d1}^2 \\ &+ C_1(A_1 + 2B_1V_{gg})v_{g1}v_{d1}^2 + B_1C_1v_{g1}^2v_{d1}^2] \end{aligned}$$

and

$$\begin{aligned} i_2 = G_0[&V_{dd}(1 + A_2V_{gg} + B_2V_{gg}^2)(1 + C_2V_{dd}) \\ &+ V_{dd}(1 + C_2V_{dd})(A_2v_{g2} + 2B_2V_{gg}v_{g2} + B_2v_{g2}^2) \\ &+ (1 + A_2V_{gg} + B_2V_{gg}^2)(1 + 2C_2V_{dd})v_{d2} \\ &+ (A_2 + 2B_2V_{gg})(1 + 2C_2V_{dd})v_{g2}v_{d2} \\ &+ B_2(1 + 2C_2V_{dd})v_{g2}^2v_{d2} + C_2(1 + A_2V_{gg} + B_2V_{gg}^2)v_{d2}^2 \\ &+ C_2(A_2 + 2B_2V_{gg})v_{g2}v_{d2}^2 + B_2C_2v_{g2}^2v_{d2}^2]. \end{aligned}$$

Here we have used the fact that the dc gate and drain voltages for both transistors are identical.

At this point it is worthwhile to consider the case of the dc drain voltage being zero, since great simplification of the two current expressions is achieved. If it is necessary to use drain bias, a parallel analysis to the one following can be made, providing similar results, with the addition of a dc term in the output current. Setting $V_{dd} = 0$,

$$\begin{aligned} i_1 = G_0[&(1 + A_1V_{gg} + B_1V_{gg}^2)v_{d1} + (A_1 + 2B_1V_{gg})v_{g1}v_{d1} \\ &+ B_1v_{g1}^2v_{d1} + C_1(1 + A_1V_{gg} + B_1V_{gg}^2)v_{d1}^2 \\ &+ C_1(A_1 + 2B_1V_{gg})v_{g1}v_{d1}^2 + B_1C_1v_{g1}^2v_{d1}^2] \end{aligned}$$

and

$$\begin{aligned} i_2 = G_0[&(1 + A_2V_{gg} + B_2V_{gg}^2)v_{d2} + (A_2 + 2B_2V_{gg})v_{g2}v_{d2} \\ &+ B_2v_{g2}^2v_{d2} + C_2(1 + A_2V_{gg} + B_2V_{gg}^2)v_{d2}^2 \\ &+ C_2(A_2 + 2B_2V_{gg})v_{g2}v_{d2}^2 + B_2C_2v_{g2}^2v_{d2}^2]. \end{aligned}$$

The load current is

$$\begin{aligned} i_0 &= i_1 + i_2 \\ &= G_0[(1 + A_1V_{gg} + B_1V_{gg}^2)v_{d1} + (A_1 + 2B_1V_{gg})v_{g1}v_{d1} \\ &\quad + B_1v_{g1}^2v_{d1} + C_1(1 + A_1V_{gg} + B_1V_{gg}^2)v_{d1}^2 \\ &\quad + C_1(A_1 + 2B_1V_{gg})v_{g1}v_{d1}^2 + B_1C_1v_{g1}^2v_{d1}^2 \\ &\quad + (1 + A_2V_{gg} + B_2V_{gg}^2)v_{d2} + (A_2 + 2B_2V_{gg})v_{g2}v_{d2} \\ &\quad + B_2v_{g2}^2v_{d2} + C_2(1 + A_2V_{gg} + B_2V_{gg}^2)v_{d2}^2 \\ &\quad + C_2(A_2 + 2B_2V_{gg})v_{g2}v_{d2}^2 + B_2C_2v_{g2}^2v_{d2}^2]. \end{aligned}$$

By appropriately balancing the potentiometers, some undersirable terms disappear, as follows: let

$$D_1 = 1 + A_1V_{gg} + B_1V_{gg}^2, \quad D_2 = 1 + A_2V_{gg} + B_2V_{gg}^2,$$
$$E_1 = A_1 + 2B_1V_{gg}, \quad E_2 = A_2 + 2B_2V_{gg}.$$

These are constants dependent only upon the transistor characteristics and the biasing point. Assume that the transformers have unity turns ratio, and therefore let

$$v_{d1} = k_de_1, \quad v_{d2} = -(1 - k_d)e_1,$$
$$v_{g1} = k_ge_2, \quad v_{g2} = -(1 - k_g)e_2.$$

where k_d and k_g are as previously defined. The expression for the load current becomes

$$\begin{aligned} i_0 &= G_0[(D_1 + B_1k_g^2e_2^2)k_de_1 + E_1k_dk_ge_1e_2 \\ &\quad + C_1D_1k_d^2e_1^2 + C_1E_1k_d^2k_ge_1^2e_2 + B_1C_1k_d^2k_g^2e_1^2e_2^2 \\ &\quad - (D_2 + B_2[1 - k_g]^2e_2^2)(1 - k_d)e_1 \\ &\quad + E_2(1 - k_d)(1 - k_g)e_1e_2 + C_2D_2(1 - k_d)^2e_1^2 \\ &\quad - C_2E_2(1 - k_d)^2(1 - k_g)e_1^2e_2 \\ &\quad + B_2C_2(1 - k_d)^2(1 - k_g)^2e_1^2e_2^2] \\ &= G_0\{[D_1k_d + B_1k_dk_g^2e_2^2 - D_2(1 - k_d) \\ &\qquad - B_2(1 - k_g)^2(1 - k_d)e_2^2]e_1 \\ &\qquad + [E_1k_dk_g + E_2(1 - k_d)(1 - k_g)]e_1e_2 \\ &\qquad + [C_1D_1k_d^2 + C_2D_2(1 - k_d)^2]e_1^2 \\ &\qquad + [C_1E_1k_d^2k_g - C_2E_2(1 - k_d)^2(1 - k_g)]e_1^2e_2 \\ &\qquad + [B_1C_1k_d^2k_g^2 + B_2C_2(1 - k_d^2)(1 - k_g^2)^2]e_1^2e_2^2\}. \end{aligned}$$

Consider the first term:

$$[D_1k_d + B_1k_dk_g^2e_2^2 - D_2(1 - k_d) - B_2(1 - k_g)^2(1 - k_d)e_2^2]e_1.$$

Factoring, this becomes

$$\left[D_1k_d\left(1 + \frac{B_1k_g^2e_2^2}{D_1}\right) - D_2(1 - k_d)\left(1 + \frac{B_2[1 - k_g]^2e_2^2}{D_2}\right)\right]e_1.$$

By appropriately setting the gate potentiometer, the following proportion may be obtained:

$$\frac{B_1k_g^2e_2^2}{D_1} = \frac{B_2[1 - k_g]^2e_2^2}{D_2},$$

or

$$k_g = \frac{\dfrac{B_2}{D_2} \pm \sqrt{\dfrac{B_1B_2}{D_1D_2}}}{\dfrac{B_2}{D_2} - \dfrac{B_1}{D_1}}. \tag{5}$$

Note that k_g is always nonnegative and the adjustment is therefore always possible. By adjusting the drain potentiometer, we obtain the relation

$$D_1k_d = D_2(1 - k_d),$$

or

$$k_d = \frac{D_2}{D_1 + D_2}. \quad (6)$$

With these adjustments, the first term disappears.

The other terms of (4), however, cannot be balanced out using the settings of (5) and (6). Other sets of potentiometer settings are required to balance out the remaining undesired terms, one particular set for each term. Using the settings (5) and (6), the expression for the output current takes the form,

$$i_0 = K_1e_1e_2 + K_2e_1^2 + K_3e_1^2e_2 + K_4e_1^2e_2^2, \quad (7)$$

where K_1, K_2, K_3 and K_4 are the appropriate constants.

Assuming that the V-I characteristics of the transistors are linear, [that is $C = 0$ in (2)], but a nonlinearity exists in the conductance characteristics, a similar analysis shows that all the distortion terms can be balanced out. With this assumption,

$$i_0 = K_1e_1e_2.$$

Assuming that the conductance characteristics are linear [that is, $B = 0$ in (3)], but a nonlinearity appears in the V-I characteristics, the analysis will show that all distortion terms but one can be cancelled out. The resulting output current is

$$i_0 = K_5e_1e_2 + K_6e_1^2.$$

The conditions for (7) were observed in the laboratory.

Correspondence

Discussion on "Information Bandwidth of Tropospheric Scatter Systems"*

As a footnote to Mr. Shaft's interesting paper,[1] it seems worth recalling an attack on the same problem by this writer and a colleague a few years ago.[2] Like Shaft, we took as a starting point the large echo formulas developed by Medhurst and Small in 1956.[3] Considering the simple two-path model, we arrived at the approximate formula

$$\frac{D}{S} = \frac{1}{2d^2}\,\tau^2\omega_\Delta\omega_m \text{ (voltage ratio)}, \quad (1)$$

for the maximum distortion/signal in the top channel. Here,

$1/d$ = instantaneous fading depth (voltage ratio),
τ = delay difference in the two paths (seconds),
ω_Δ = rms frequency deviation due to the multichannel frequency division multiplex signal (radians per second), and
ω_m is the maximum modulating frequency (radians per second).

The error in this formula is as follows:

Instantaneous fading depth	Excess of distortion given by approximate expression (1) over distortion calculated from exact formula
10 db	2.1 db
20	0.6
30	0.2

The formula might be used to shorten quite considerably the labor of deriving curves such as those of Shaft's Figs. 3–6.

Shaft remarks that the analytical difficulties involved in the two-path model might discourage investigation of a multipath model. In Medhurst and Hodgkinson[2] we did find it possible to arrive at a surprisingly simple approximate expression for the worst intermodulation distortion due to multipath fading. For top channel, this is

$$\frac{D}{S} = \frac{1}{2d^2}\left(\sum_{n=1}^{N} r_n\tau_n\right)^2\omega_\Delta\omega_m \text{ (voltage ratio)}, \quad (2)$$

where N is the number of secondary paths, r_n is the amplitude of the signal transmitted over the nth secondary path relative to that over the main path (voltage ratio) and τ_n is the delay difference in the nth path. This formula should be of the same order of accuracy as (1).

The difficulty in drawing conclusions from (2) is that so little experimental information is available concerning the multipath structure of frequency-selective fading. In Medhurst and Hodgkinson[2] it was shown, for the one fade analyzed into its echo structure that we could locate in the literature,[4] that the intermodulation distortion predicted by the present theory would be quite severe. There would seem to be an excellent case for an extended experimental investigation on the lines of the one initiated by Kaylor.[4]

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[1] P. D. Shaft, "Information bandwidth of tropospheric scatter systems," IRE Trans. on Communications Systems, vol. CS-9, pp. 280–287; September, 1961.

[2] R. G. Medhurst and M. Hodgkinson, "Intermodulation distortion due to fading in frequency-modulation frequency-division multiplex trunk radio systems," IEE Monograph No. 240R; May, 1957 (reprinted in *Proc. IEE*, pt. C, vol. 104, pp. 475–480; September, 1957).

[3] R. G. Medhurst and G. F. Small, "An extended analysis of echo distortion in the F.M. transmission of frequency-division multiplex," *Proc. IEE*, vol. 103, pt. B, pp. 190–198; March, 1956.

[4] R. L. Kaylor, "A statistical study of selective fading of super-high frequency radio signals," *Bell Sys. Tech. J.*, vol. 32, pp. 1187–1202; September, 1953.